

CHEMICAL CHANGES – REACTIONS OF ACIDS

KEY VOCABULARY

Acid: produces H^+ ions in solution.

Base: neutralises an acid (metal oxides/hydroxides).

Salt: formed when the H^+ of an acid is replaced by a metal.

Neutralisation: acid + base $\rightarrow$ salt + water.

COMMON MISCONCEPTIONS

~~Acid + metal makes water.~~

✓ Acid + metal $\rightarrow$ salt + hydrogen (no water).

~~All salts are sodium chloride.~~

✓ A salt is any metal + acid-ion compound.

~~Carbonates give off oxygen with acid.~~

✓ Carbonates give off carbon dioxide.

GENERAL EQUATIONS

acid + metal $\rightarrow$ salt + hydrogen

acid + metal oxide $\rightarrow$ salt + water

acid + metal hydroxide $\rightarrow$ salt + water

acid + carbonate $\rightarrow$ salt + water + CO_2

$\text{HCl} \rightarrow$ chloride · $\text{H}_2\text{SO}_4 \rightarrow$ sulfate · $\text{HNO}_3 \rightarrow$ nitrate

1 ACID + METAL

Name the two products of acid + metal. [2 marks]

2 NAME THE SALT

Name the salt made from hydrochloric acid and zinc. [1 mark]

3 CARBONATE TEST

Explain how you would show that a carbonate gives off CO_2 with acid. [2 marks]

4 WRITE AN EQUATION

Write a word equation for sulfuric acid + copper oxide. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Acid + metal oxide $\rightarrow$ ____ + ____.

water

hydrogen

salt

oxygen

6 PREDICT PRODUCTS

Predict the products of nitric acid + calcium carbonate. [3 marks]